

AUTHORITATIVE SDLC & GOVERNANCE BLUEPRINT

Rankuno AI Engine - Master System Architecture

Google Antigravity Agentic Stack, Mandatory 8-Step SDLC Protocol, & Directory Standards

1. Google Antigravity Agentic Stack Mapping

Layer	Component	Google Antigravity Equivalent	Rankuno Implementation Role
Model	Reasoning Engine	Gemini 3.6 Flash / Pro	Reasoning, architectural planning, and code generation.
Agent	Decision Maker	Antigravity Execution Loop	Primary orchestrator executing Plan → Act → Observe → Reason loop.
Subagents	Specialized Delegates	invoke_subagent	Background workers for research, audits, or isolated test suites.
Skills	Procedural Guides	skills/<name>/SKILL.md	Reusable procedural guides (antigravity-sdlc-flow, seo-engine-guide).
Tools	Native Capabilities	Atomic capabilities	read_file, write_file, run_command, grep_search.
MCP	External Connectors	Model Context Protocol	Connectors for Search Console, Google Ads API, SERP scrapers.

2. Mandatory 8-Step SDLC Execution Protocol

- Step 1: Investigation & Requirement Discovery:** Analyze problem, domain rules, and verify code.
- Step 2: System Architecture & Data Schema Blueprint:** Define Pydantic models and component boundaries.
- Step 3: Human-in-the-Loop (HITL) Review Checkpoint:** Present architecture to operator for explicit approval.
- Step 4: Reasoning-Driven Implementation Plan:** Detail line-by-line file changes and logic.
- Step 5: Security, Rate-Limit & Edge-Case Audit:** Audit for anti-bot blocks, rate limits, API spend, breaking changes.
- Step 6: Step-by-Step Modular Implementation:** Write clean, strictly typed, modular code.
- Step 7: Automated Unit Testing & Verification:** Execute unit test suites (pytest); never declare success without passing tests.

8. **Step 8: README & Architecture Drift Audit:** Keep documentation in sync with codebase changes immediately.

3. Phase Architecture Index

- **Phase 1: Multi-Dimensional Page Classification Engine** → docs/ARCHITECTURE_PHASE1_CLASSIFICATION.md
- **Phase 2: Technical SEO Crawling & Audit Engine** → [In Discovery]
- **Phase 3: Keyword Intelligence & Vector Clustering Engine** → [In Discovery]
- **Phase 4: Paid Search (PPC) Campaign & Ad Engine** → [In Discovery]
- **Phase 5: Market Research & Competitor Intelligence Engine** → [In Discovery]

4. Master Module Directory Layout

```
src/
├── core/                                # Core Agentic Utilities & API Guardrails
│   ├── base_tool.py                    # Pydantic Tool Interface base class
│   ├── rate_limiter.py                 # API Rate Limiter & Cost Guard
│   └── logger.py                       # Structured Execution Audit Logging
├── modules/
│   ├── seo/                            # SEO Engine Module
│   │   ├── classification/             # Phase 1: Multi-Dimensional Page Classifier
│   │   ├── audit/                     # Technical SEO Crawler & HTML Parser
│   │   ├── clustering/                 # Semantic Keyword Embeddings & Intent Clustering
│   │   └── rank_tracker/               # GSC API Rank Tracking Synthesizer
│   ├── ppc/                           # PPC Campaign Engine Module
│   └── research/                       # Market Research & SERP Scraper Module
├── integrations/                       # External API Wrappers
│   ├── gsc_client.py                  # Google Search Console API Wrapper
│   ├── google_ads_client.py           # Google Ads API Wrapper
│   └── serp_client.py                 # SERP Data Scraper & API Wrapper
```